

Keerti Yadav

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Summary

Ph.D. geographer and geospatial data scientist with 3+ years of experience spanning spatial data analysis, transit and mobility data systems, remote sensing/Earth observation, and quasi-experimental causal inference for urban infrastructure and transport policy. Doctoral research applies staggered Difference-in-Differences (Callaway-Sant'Anna) methodology to estimate the heterogeneous regional effects of high-speed rail across 31 South Korean cities and nine transit corridors (2008-2020), building and validating a multi-source panel dataset that integrates administrative records, road-network and infrastructure variables, and satellite-derived (Landsat/Sentinel-2 NDBI) land-use change. Applied Earth observation work extends to drought and heatwave risk modeling and land-use vulnerability assessment across South Korea and India. Proficient in Python and R for spatial data pipelines and analytical modeling, and in ArcGIS, QGIS, Google Earth Engine, OSMnx, and the Kakao Maps API for GIS, remote sensing, and transit-network analysis.

Education

PhD	Kongju National University, South Korea , Geography • Thesis: Heterogeneous Regional Effects of High-Speed Rail on Small and Medium-Sized Cities in South Korea: Focusing on Staggered Difference-in-Differences Analysis (2008-2020)	Mar 2022 – Feb 2026
	Busan University of Foreign Studies, South Korea , Language Training Program	Jan 2021 – Feb 2022
MA	Jawaharlal Nehru University, India , Korean Studies, School of Language, Literature and Cultural Studies	Aug 2019 – May 2021
BA	Jawaharlal Nehru University, India , Korean Studies, School of Language, Literature and Cultural Studies	Aug 2016 – May 2019

Experience

Kongju National University, South Korea , Ph.D. Researcher - Regional Development & Urban Transport	Mar 2022 – Feb 2026 4 years
- Isolated heterogeneous regional treatment effects of high-speed rail across cities (2008-2020) by validating a quasi-experimental design. - Delivered an analysis-ready, multi-source panel dataset spanning 31 cities and 13 years by resolving data-source misalignments, missing-value gaps, and administrative boundary/name inconsistencies, using custom R- and Python-based crosswalk and geocoding routines. - Quantified land-use and built-up change across cities by integrating Landsat/Sentinel-2 NDBI satellite imagery with socioeconomic panel data in R, ArcGIS, and Google Earth Engine, enabling city-level geospatial impact analysis at scale. - Corrected geographic boundary errors and closed data gaps across the panel dataset by cross-checking satellite-derived land-use change against socioeconomic and infrastructure records in ArcGIS, QGIS, and Google Earth Engine, aligning results with local administrative policy. - Reduced manual data-integration effort on recurring road-network updates by building automated cross-join pipelines in Python and R for expressway and national-highway variables, cutting ingestion and quality-check time across the dataset. - Built reusable working knowledge of Korean public and geospatial datasets by using OSMnx to extract and analyze street-network data and the Kakao Maps API to plot and cross-check Korea-specific spatial resources. - Produced actionable, country-specific transit and mobility recommendations for India and South Korea by synthesizing policy documents, spatial data, and governance case studies with local governance bodies and academic partners into a cross-national research framework. - Sustained reproducibility across a multi-year, multi-partner project by documenting research methodology and data pipelines throughout, supporting consistent process compliance for all collaborators.	
Kongju National University, South Korea , Research Assistant - Mapping & Spatial Data	Jan 2023 – Dec 2024 2 years
Maintained data integrity for university mapping projects by supporting collection, cleaning, and validation processes in coordination with faculty and external data providers.	
International forums & institutions, South Korea & India , Korean-English Translator & Cultural Liaison / International Program Coordinator	Jan 2020 – present 6 years 8 months
- Enabled clear communication across 5+ major cross-border programs (2023-2026) by serving as trusted translator and cultural liaison between Korean institutions and international/Indian delegations, including the 47th International Exhibition on Environmental Technology & Green Energy (Seoul, 2026), International Forum on One Korea (Seoul, 2024), Korea International Model United Nations (Seoul, 2024), Global Peace Leadership Corps (Seoul, 2024), and the Andong Cultural Exchange (Andong, 2023). - Delivered a 10-day international exchange program for HUFS students (Delhi, 2020) by organizing and leading end-to-end travel, scheduling, and on-the-ground coordination.	

Publications

Impact of High-Speed Rail on Urban Regions in South Korea: A Systematic Literature Review	Jan 2025
Y. Keerti, J. H. Jin (The Geographical Journal of Korea)	

A Systematic Approach for Restoring Sustainability in the Hilly Regions of Northeast

India: The Zabö Conceptual Framework

K. Yadav, V. Choudhary

(GeoJournal (under second-stage review))

Training-Free Vision-Language Models for Near-Surface GPR Radargram Interpretation:

A Benchmark of Subsurface Anomaly Detection and Reliability

K. Baluch, M. F. Nasir, K. Yadav, S. A. Qaisrani, J. Wang

(Journal of Geophysical Research: Solid Earth (under second-stage review))

Conference Presentations

"Climate Change Vulnerability of Traditional Land Use Practices in Zabö Agriculture, India" - KAGIS Fall Conference & IAGI International Symposium, 2025

"Exploring Sustainability through existing and potential GIAHS (Globally Important Agricultural Heritage Systems)" - National Conference on Sustainable Rural Development, October 2023

"Human Resource: Success Key of South Korea" - 7th AKS Young Scholars' International Webinar on Korean Studies, JNU New Delhi, June 2020

Technical Skills

Geospatial & Spatial Analysis: ArcGIS, QGIS, spatial data auditing and QA, boundary/attribute validation, multi-source dataset integration

Remote Sensing & Earth Observation: Google Earth Engine, Landsat/Sentinel-2 NDBI land-use and land-cover change analysis, satellite-derived variable integration

Transit & Mobility Data: OSMnx street-network extraction, HSR/transit corridor panel datasets, transit accessibility analysis

Causal Inference & Econometric model: Staggered Difference-in-Differences (Callaway-Sant'Anna), panel data analysis, quasi-experimental design

Programming & Data Pipelines: Python, R, automated data ingestion and quality-control pipelines

Professional Associations

American Association of Geographers

Economic Geography Research Group

Quantitative Methods Research Group

Transport Geography Research Group

Urban Geography Research Group

Honors Awards Service

Government of Korea (GKS/Global Korea Scholarship) recipient

AKS International Young Scholars' Seminar

Certification: Test of Proficiency in Korean (TOPIK)

Presenter, Korea International Model UN 2024, DISEC Committee, Seoul

Languages

English: Native / Bilingual

Hindi: Professional Working Proficiency

Korean: Professional Working Proficiency